Diamond & Related Materials 91 (2019) 255–260

Contents lists available at ScienceDirect

Diamond & Related Materials

journal homepage: www.elsevier.com/locate/diamond

One-pot millisecond preparation of quench-resistant solid-state fluorescence carbon dots toward an efficient lubrication additive

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ARTICLE INFO

Keywords:
Carbon dots
Solid state fluorescent
Tribological properties
Gaseous detonation

ABSTRACT

Currently, the investigation on preparation, photoluminescence (PL) mechanism and application of solid state fluorescent CDs (SSFCDs) is still in the preliminary stage. It is particularly urgent to research on this issue. Herein, a one-pot gaseous detonation method is employed to directly synthesize SSFCDs within milliseconds using citric acid (CA) and urea as precursors for the first time. The SSFCDs exhibit dependent emission with the excitation/emission maximum of 370/452 nm. On the basis of studying the structural and optical characteristics of the SSFCDs and thermal annealing SSFCDs (A-CDs) as well as the properties of gaseous detonation, we speculate the PL mechanism and formation mechanism of SSFCDs. More importantly, the prepared SSFCDs without any purification as additives for poly(ethylene glycol) (PEG) obtain a super-low mean friction coefficient of 0.024 and wear scar diameter of 0.57 mm under 600 N at optimal concentration of 0.5 wt%, reducing by 60.7% and 19.7% respectively in comparison with that of pure PEG. The SSFCDs-based additives reported here display excellent friction-reducing and antiwear performance for PEG.

1. Introduction

Owing to their good stability, water solubility, low toxicity, superior resistance to photobleaching, and size effect, carbon dots (CDs) have attracted intensive research interests in bioimaging [1,2], sensors [3–5], optoelectronic devices [6,7], and lubricant additives [8–10]. However, one of the major drawbacks of CDs is self-quenching in solid state because of aggregation-induced quenching (AIQ) effects, limiting the wide use of CDs [11–13]. To overcome this drawback, encapsulation methods to embed CDs in solid matrices, such as starch [14], silica xerogel [15], polymers [16–18] and trisodium citrate [19], have been carried out to realize solid-state fluorescence. However, these treatments essentially fail to conquer the AIQ effects because the matrices like water act as dispersants.

Recently, a few studies about CDs circumventing AIQ effects have been reported by developing newly designed structure and emission center. In detail, Liu's group [11] and Yang's group [12] use polymer precursors as steric hindrance chains, controlling interparticle spacing to realize solid state fluorescent CDs (SSFCDs). Zeng's group [20] successfully synthesizes SSFCDs by designing pure surface states. Yang's group [21], Zhou' group [22] and Kim's group [23] employ molecule precursors to prepare SSFCDs through control of the carbonization degree to avoid the formation of strong supramolecular interaction. Generally, the microwave method is prevalent to the prepare SSFCDs because of milder reaction conditions, time-saving process, and higher chemical yields [21,22,24]. However, the limitations of microwave synthesis cannot be neglected, such as a little production and complicated post-treatment [25]. It is desirable to develop a facile, rapid and efficient synthetic approach for SSFCDs.

Gaseous detonation, generating detonation wave induced by combustible gas and oxygen to causing a complex and fast series of chemical reactions in precursors, has been successfully applied to the synthesis of metal oxides and carbon composites, such as nano-TiO_2 [26], carbon nanotubes [27], carbon onions [28] and graphene quantum dots [25]. Solid-state nanomaterials on a large scale within milliseconds using untreated precursors can be directly obtained by the gaseous detonation approach. For all we know, there is no report about the synthesis of SSFCDs by gaseous detonation.

In this study, we report a one-pot and facile gaseous detonation method to directly synthesize SSFCDs within milliseconds using low-cost citric acid (CA) and urea as precursors. Through studying the structural and optical characteristics of SSFCDs and thermal annealing SSFCDs (A-CDs), it is possible to find out the PL mechanism of SSFCDs. Furthermore, based on various possible reaction pathways and the properties of gaseous detonation, we speculate the formation mechanism of SSFCDs. More importantly, the prepared SSFCDs without

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https://doi.org/10.1016/j.diamond.2018.12.003
Received 21 October 2018; Received in revised form 4 December 2018; Accepted 5 December 2018
Available online 06 December 2018
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any purification excellent friction-reducing and antiwear properties as additives for poly(ethylene glycol) (PEG), obtaining a super-low mean friction coefficient of 0.024 and wear scar diameter of 0.57 mm under 600 N at optimal concentration of 0.5 wt%. The SSFCDs-based additives reported here display excellent friction-reducing and antiwear performance for PEG.

2. Experimental

2.1. Materials

Citric acid (CA, AR grade), urea (AR grade) and PEG with an average molecular weight of 380–420 g mol^{-1} (AR grade) were obtained from Tianjin Kermel Chemical Regent Co., Ltd. (China). Hydrogen and oxygen (> 99.999%) were purchased from Dalian Special Gases Co., Ltd. (China). All chemicals were used as received without further purification.

2.2. Instrumentation

High resolution transmission electron microscopy (HRTEM) images were taken with a Tecnai F20 transmission electron microscope (FEI, USA). Atomic force microscopy (AFM) images were recorded using a MultiMode3D atomic force microscopy (Veeco, USA). Raman spectra were confirmed by a DXR microscope Raman spectrometer (Thermo Fisher, USA). Fourier transform infrared (FT-IR) spectra were characterized using a Nicolet-6700 FTIR spectrophotometer (Thermo, USA). UV–Vis absorption spectra were acquired using a Lambda 750 UV–Vis-NIR spectrophotometer (Perkin Elmer, USA). Photoluminescence (PL) spectra were obtained by an F-7000 fluorescence spectrophotometer (Hitachi High Technologies, Japan). The friction performance was conducted on a MMW-1A microcomputer control friction and wear tester (Jinan Yihua Tribology Testing Technology Co., Ltd. China) through a four-ball model under the following conditions: rotating rate, 1200 rpm; duration, 60 min; load, 600 N; and temperature, 75 °C. The wear scar diameters were measured by an OLS4000 confocal laser scanning microscopy (Olympus, Japan).

2.3. Synthesis of SSFCDs

The preparation of SSFCDs was as follows. CA (4 g) and urea (4 g) were blended and directly put in the self-made detonation tube. Secondly, the detonation tube was vacuumized and heated to 85 °C. Then, hydrogen and oxygen in the mole ratio of 2:1 as the explosion source were pumped into the tube, respectively. Finally, a rapid detonation reaction was caused by igniting the gas mixture. Under the effect of detonation, the CA and urea were converted into viscous dark-brown SSFCDs (1.3 g, shown in Fig. 1d) within 3–5 ms.

2.4. Preparation of thermal annealing SSFCDs (A-CDs)

The A-CDs were prepared using a tube furnace to heat the SSFCDs put in a quartz boat at 300 °C for 15 min under nitrogen atmosphere.

2.5. Preparation of the lubricant suspension

The lubricant suspension was prepared by adding certain amounts of the untreated SSFCDs into PEG under ultrasonic agitation for 15 min at room temperature.

3. Results and discussion

3.1. Characterizations of SSFCDs

Figs. 1a and S1a show the morphology of the SSFCDs as characterized by HRTEM. It is observed that most of the SSFCDs are well dispersed and display spherical particle shape with a size distribution within the range of 4.0 to 7.6 nm and an average diameter of ca. 5.7 nm. Besides, these particles are observed to be amorphous carbon particles without any lattices, indicating that no graphite structure formed during the gaseous detonation process (Fig. S1a). The AFM image illustrates an average topographical height of 3.5 nm (Fig. S1c and d). No obvious D or G bands are detected in the Raman spectrum (Fig. S1b), further suggesting the amorphous nature of the SSFCDs [29]. FT-IR spectrum (Fig. 1b) indicates that the SSFCDs possess various functional groups. The main absorption peaks at 3420 and 3215 cm^{-1} are attributed to the stretching vibration of O—H and N—H groups [1], endowing the SSFCDs with high water-solubility [30]. The strong peak of 1572 cm^{-1} is assigned to bending vibrations of N—H [21,29], while the peaks at 1396 and 1078 cm^{-1} belong to bending vibration of C—O and stretching peak of C—O—C, respectively [24]. In addition, the weak peak at 1655 cm^{-1} proves the existence of C═O [22].

The SSFCDs aqueous solution displays yellow color exposure to sunlight and emits a blue fluorescence excited with 365 nm UV light (Fig. 1c). Its UV–Vis absorption spectrum (Fig. 1c) shows an intense absorption at 335 nm, assigned to the typical n–π* transition of nitrogen or oxygen-based groups [1,31]. The SSFCDs aqueous solution with the concentration of 1.0 mg mL^{-1} exhibits excitation-dependent feature with excitation/emission maximum of 350/430 nm (Fig. S1e). However, the SSFCDs are dark-brown under daylight and show a bright blue fluorescence upon exposure to UV irradiation at 365 nm (Fig. 1d). Compared with its aqueous solution, the SSFCDs also show excitation dependence, but a slight red-shift excitation/emission maximum of 370/452 nm. When the excitation wavelength varies from 340 to 400 nm, the emission wavelength is red-shift correspondingly from 435 to 464 nm. Interestingly, the emission maximum peak and its intensity of SSFCDs aqueous solution are dependent on the concentrations (Fig. S1f). With the concentrations increasing, the emission maximum displays a slight red shift while its intensity significantly enhances. Besides, the prepared SSFCDs possess stronger emission maximum than the SSFCDs aqueous solution, which is beneficial for wider use (Fig. S1f).

3.2. PL mechanism of SSFCDs

Currently, the investigation for PL mechanism of SSFCDs is still in the preliminary stage [32]. Zeng's group has revealed that the PL origin can be summarized into the surface states [20]. In detail, various surface groups on the surface of CDs, acting as electron donors and acceptors, form π-π stacking and induce AIQ effect. On the contrary, the CDs with relative pure surface state can realize solid-state fluorescence emission. However, the proposed model cannot account for the SSFCDs with multiple surface states. Liu's group has demonstrated that surface polymer chains can prevent the graphitizing cores from π–π interactions and resist AIQ effect [11]. The drawback of this assumption is only appropriate for SSFCDs with graphite structure and polymer chains. More recently, Yang's group has attributed the PL origin of SSFCDs to the crosslinked polymer chains with net structure and the unique fluorescence centers (named sub-fluorophores) [21], which is helpful for solid-state fluorescent polymer CDs.

Despite considerable efforts focused on the emission origin of SSFCDs, it still remains unclear. By investigation on the structural and optical characteristics of SSFCDs and A-CDs, we attempt to explore PL mechanism of SSFCDs; that is the various fluorophore molecules, including multitudes of nitrogen- and oxygen-containing functional groups as well as isolated within crosslinked polymer chains, dominate quench-resistant solid-state fluorescence, which is supported by following evidences (Fig. 2a). First of all, traditionally, there are three different types of emission centers in CDs, i.e., molecular states, aromatic domain states, and carbon-core states [33–35]. HRTEM images and Raman spectrum of SSFCDs suggest there are few extended conjugation structures, such as polycyclic aromatic hydrocarbons (PAHs)

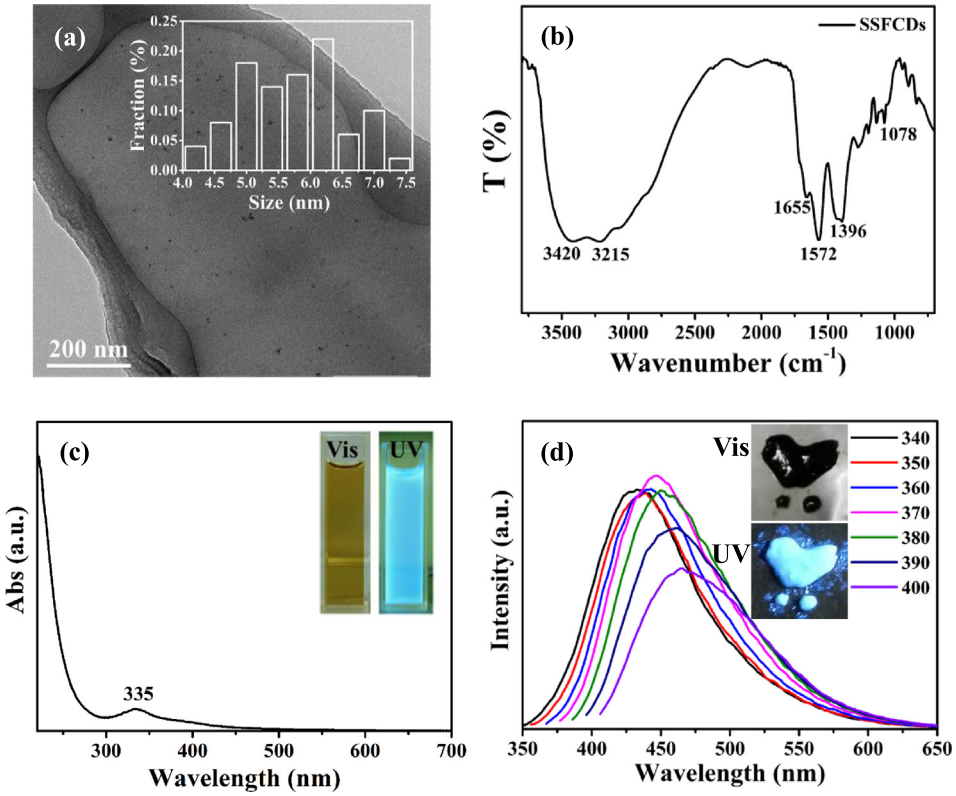


Fig. 1. Morphology and optical properties of SSFCDs: (a) HRTEM image (with size distribution, inset); (b) FT-IR spectrum; (c) UV–Vis absorption spectrum of SSFCDs aqueous solution (with optical photographs taken under sunlight and a 365 nm UV light, inset); (d) PL emission spectra (with photographs under visible light and under a 365 nm UV light, inset).

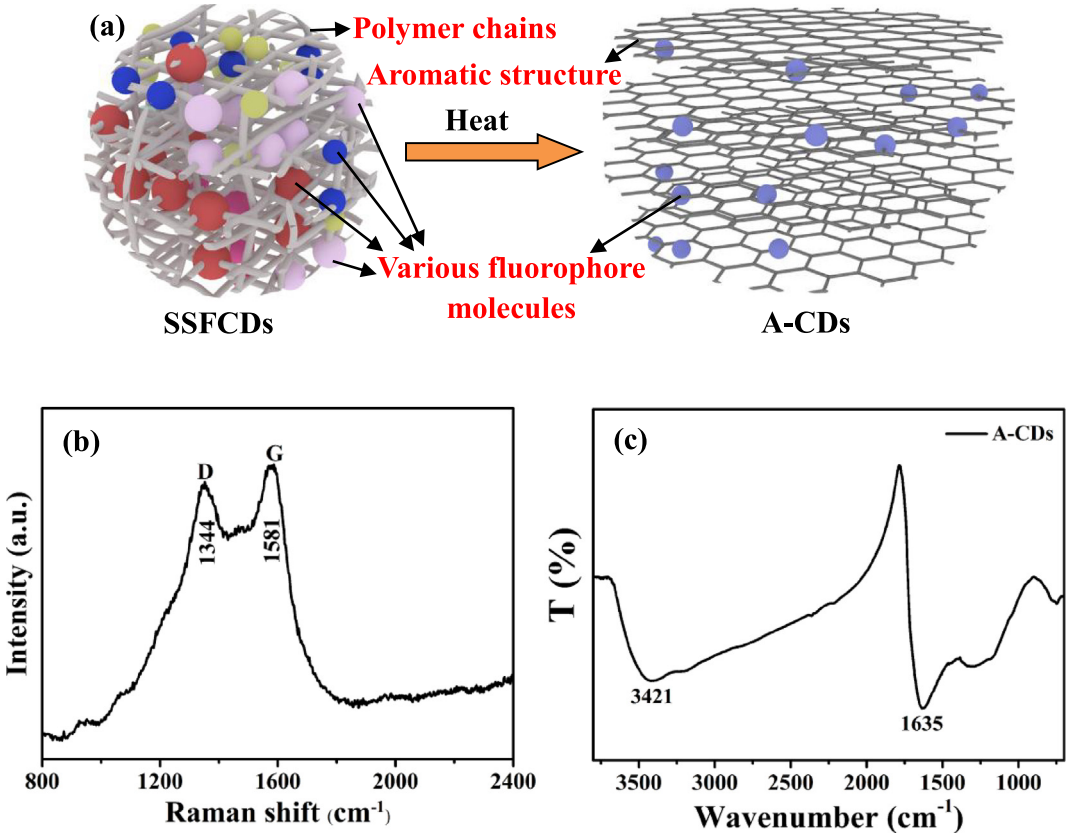


Fig. 2. (a) The sketches of SSFCDs and A-CDs; (b) Raman and (c) FT-IR spectrum of A-CDs.

and carbon-cores. No characteristic peaks of aromatic structure appear in UV–Vis spectrum, further demonstrated that the fluorescence is unlikely to result from aromatic domain states and carbon-core states. Therefore, the PL origin of SSFCDs is only from fluorophore molecular states, and the crosslinked polymer chains, acting as steric hindrance chains, encapsulate the fluorophore molecules and control their interparticle spacing to realize solid state fluorescent. Moreover, the emission maximum (452 nm) of the obtained SSFCDs has a slight red-shift of 22 nm from that of dilute aqueous solution (430 nm), much smaller than previous reports [11,21] and in agreement with organic molecules [36]. It is also believed that the smaller red-shift demonstrating a little resonance energy transfer or re-absorption [11,37], is ascribed to the fact that polymer chains in SSFCDs provide fluorophore molecules efficient separation. It should be pointed out that the distinct PL characteristics of SSFCDs including excitation wavelength dependent PL and broad peak width, inconsistent with molecular fluorescence features, may be attributed to coexistence of various fluorophore molecules formed during gaseous detonation and serving as different excited states.

The properties of A-CDs are used to further validate PL mechanism. SSFCDs went through further dehydration and carbonization to form A-CDs under thermal treatment. A-CDs, becoming black powder without fluorescence under a 365 nm UV light, display excitation-independent feature with a large blue shift excitation/emission maximum at 240/385 nm (Fig. S2). The Raman spectrum of A-CDs (Fig. 2b) exhibits the typical D band at 1344 cm^{-1} and G band at 1581 cm^{-1}, ascribed to disordered sp^3 and the crystalline sp^2, respectively [38]. The I_D/I_G ratio is 0.94, suggesting that A-CDs have a higher carbonization degree than SSFCDs [39]. The FT-IR spectrum of A-CDs (Fig. 2c) displays strong aromatic C=C vibration at 1635 cm^{-1} [40] and stretching vibration peaks of O−H at 3421 cm^{-1} [1,41], indicating that A-CDs have few nitrogen-containing functional groups but plenty of aromatic structures. As a result, A-CDs fail to dissolve in water or alcohol. All the results above coincide with our hypothesis of PL mechanism. Due to losing almost all nitrogen-containing groups and including only fluorescence molecules with OH groups, A-CDs cannot show blue fluorescence and its excitation-independent feature with the excitation/emission maximum at 240/385 nm is put down to the fluorescence molecule containing only OH group (Fig. 2a). It should be noted that despite further dehydration and carbonization under thermal treatment contributing to the fact that most of the fluorophore molecules and polymer chains turn into aromatic conjugation structure, there is little fluorescence from aromatic domain states or carbon-core states. The reason is that after annealing, sp^2 subdomains, serving as PL origin, are apt to percolate and interconnect, leading to fluorescence quenching [25,42,43].

3.3. Synthesis mechanism of SSFCDs

Due to the difficulties in tracking and identifying the complicated reactions, the formation mechanism of CDs is usually not employed [44]. Here, on the basis of various possible reaction pathways and the characteristics of gaseous detonation, a possible synthesis mechanism is schematically explained in Fig. 3. The classical Zeldovich-von Neumann-Döring (ZND) model of detonation composed of three zones (reactants, reactions, and products) is used to explain the formation of SSFCDs. In the reaction zone, hydrogen reacts with oxygen, generating a mass of energy and leading to the results that the temperature and pressure are higher than 1500 K and 3 MPa [28]. However, it has been proven that CDs with an average diameter of ca. 5 nm are inclined to be derived at 250–300 °C [30,35]. Thus, it is difficult to form CA-based CDs under high temperature. CA and urea, as active substances under thermal treatment, partly pyrolysis into H_2O, CO_2/CO and NH_3 as proven by final reactant gas, causing a low production yield. Thus, there are few SSFCDs formed in the reaction zone. In the products zone, the temperature and pressure have a sharp decline [27]. As a result,

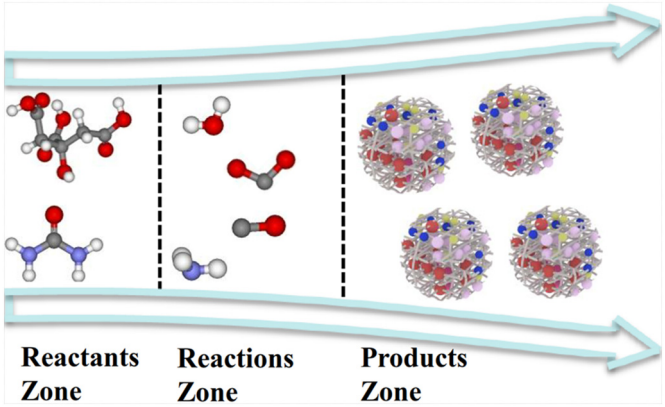


Fig. 3. The schematic representation of the formation of SSFCDs (the arrows represent the direction of SSFCDs formation).

many reaction types are possible, which is more conducive to formation of SSFCDs. Some CA and urea tend to decompose into multiple products [24], giving rise to the formation of various fluorescent molecules. Meanwhile, it is believed that CA and urea undergo a sequential dehydration and polymerization process, leading to formation of polymer structure. Since the whole reaction ends in 3–5 ms [28], the carbon condensation fails to proceed continuously. Therefore, it is hard to form the crystalline cores of CDs as confirmed by HRTEM images. If the heating continues, aromatic structures will be generated with further dehydration and carbonization as proven by A-CDs. Finally, all the intermediate products crosslink to form SSFCDs, maintaining the fluorescent molecule within polymer structure acting as polymer chains.

3.4. Tribological properties of SSFCDs

The SSFCDs with multitudes of functional groups are expected to possess good dispersion stability in PEG, which is one of the key prerequisites as nano-lubrication additives [9]. Fig. 4a illustrates photographs of SSFCDs/PEG suspensions with different concentrations under daylight and UV light after standing for one month. Neither visible precipitation nor any cloud was found in these photographs, suggesting favorable dispersion stability of SSFCDs in PEG. To evaluate the tribological performance of SSFCDs-based additives, c wt% SSFCDs/PEG (where c = 0.0, 0.1, 0.3, 0.5, 0.8, 1.0) suspensions as lubricants are tested under the load of 600 N. Compared with that of pure PEG, the wear scar diameter of SSFCDs reduces by 19.7% when the concentration is 0.5 wt% (Fig. 4b). The corresponding images of wear scar are displayed in Fig. S3. Meanwhile, with the SSFCDs added to PEG, the mean friction coefficient decreases obviously, and it reaches 0.024 at concentration of 0.5 wt%, which reduces by 60.7% in comparison with that of pure PEG. The SSFCDs-based additives reported here display excellent friction-reducing and antiwear performance for PEG [9,45,46]. The reason can be explained that the solid-state CDs possess abundant functional groups which are helpful for adsorbing and decomposing at a rubbing surface and forming a protective film which minimizes metal-to-metal contact [8].

4. Conclusions

In conclusion, SSFCDs are prepared via a rapid one-pot gaseous detonation approach using CA and urea as precursors. On the basis of studying the structural and optical characteristics of the SSFCDs and the A-CDs, the PL mechanism may be explained that the various fluorophore molecules, including multitudes of nitrogen- and oxygen-containing functional groups as well as isolated within crosslinked polymer chains, dominate quench-resistant solid-state fluorescence. Furthermore, by combining various possible reaction pathways and the

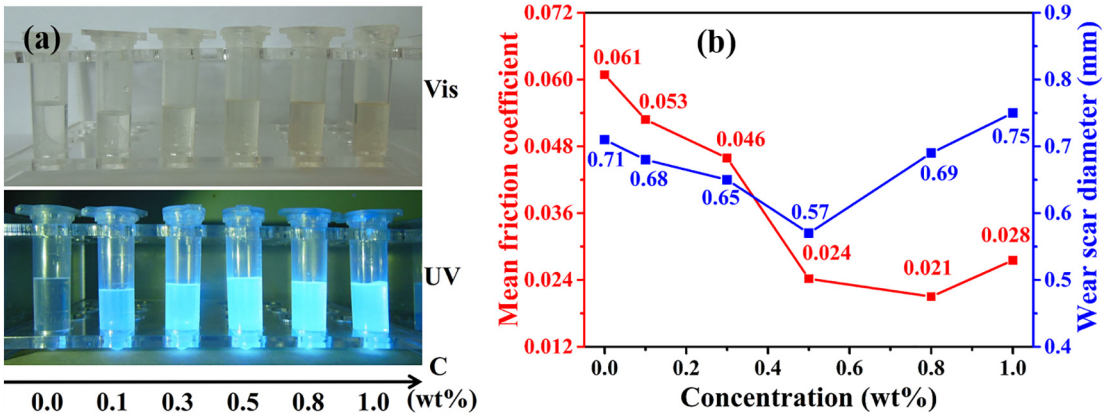


Fig. 4. (a) The photographs of SSFCDs/PEG suspensions with different concentrations under daylight and UV light; (b) the mean friction coefficient and wear scar diameter of SSFCDs at various concentrations.

properties of gaseous detonation, we speculate the formation mechanism of the SSFCDs. More importantly, the prepared SSFCDs without any purification possess excellent friction-reducing and anti-wear properties as additives for PEG.

Acknowledgments

The authors appreciate financial support from the National Natural Science Foundation of China (11672068, 10872044 and 11672067).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.diamond.2018.12.003.

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